



Methods

Preview

1. Data: acoustic/articulatory data from published datasets
- 2. Distinctiveness measurement in acoustics & articulation**
3. Bigram surprisal calculation
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target



USC



Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- The matrices were extracted through self-supervised autoencoders.
- Acoustic/articulatory data → encoder → latent space → decoder → reconstructed data

